Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Fundamental Study of Smoldering Combustion in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	03/26/1998	**End Date:**	07/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	note - date changed from 07/01/2003		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	Smolder experiments on ground-based and space-based facilities are conducted, and theoretical models of the process developed, to increase the understanding and prediction of smoldering combustion under normal and microgravity conditions. The experiments are conducted with polyurethane foam as fuel with mixtures of oxygen/nitrogen as oxidizer. Temperature at several locations of the sample are measured with thermocouples; and the resulting temperature histories are used to obtain the smolder propagation velocity and smolder reaction temperature, as a function of the oxidizer flow velocity and oxygen concentration. These data are complemented with an Ultrasound Imaging System (UIS) that measures the permeability of the foam and char, and through it provides information about the location of the smolder front, and the level of fuel consumption by the smolder reaction. The experimental results are used to verify and improve the theoretical models of smolder combustion. Experiments are also conducted to observe the potential transition from smoldering to gas flaming, and to determine the conditions and mechanisms leading to this transition.		

Task Description:	The overall project objective is to increase the fundamental understanding and the prediction of smoldering combustion under normal and microgravity conditions. This in turn will help the prevention and control of smolder originated fires both, in normal gravity and in space-based environments. The specific objectives are to determine the smolder characteristics of a polymeric porous combustible material (polyurethane foam), in a quiescent or convective oxidizing environment, both at normal and microgravity Smolder is important as both a fundamental combustion problem, i.e., the propagation of a heterogeneous, non-flaming, surface combustion reaction through a porous combustible material, and as a fire safety problem, i.e., the production of toxic compounds and the potential initiation of a fire through the transition to flaming. There is a need for further understanding of the problem for better prediction and control of smolder in both normal gravity and microgravity. Furthermore, microgravity introduces additional questions about the transport of mass to and from the reaction, and the transfer of heat from the reaction, that must be resolved in order to predict smolder behavior in a space-based environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/15/2026)